2026 BROCK AI DAY · EXPLAINABLE AI / DEEP LEARNING

Are language models converging on shared structures?

Mechanistic interpretability, the linear representation hypothesis, and the platonic representation hypothesis

James R. A. Henry · Computer Science, Brock University

MECHANISTIC INTERPRETABILITY

Reading a model from the inside

FROM THE OUTSIDE



Judge a model by what it says.

FROM THE INSIDE

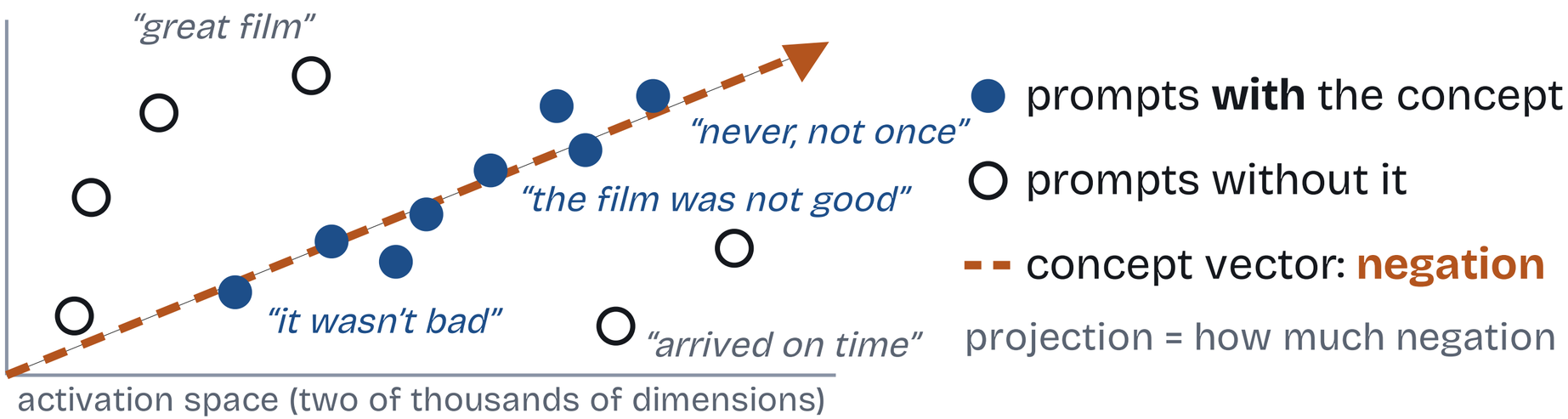


Read what it represents, and how.

It needs the model's weights, so it is only possible with open models.

CONCEPTS ARE VECTORS: THE LINEAR REPRESENTATION HYPOTHESIS

A concept is a direction



A model stores each concept it learns, such as negation, as a vector in its activation space. The more strongly a prompt expresses the concept, the further it lands along that vector.

Park, K., Choe, Y. J., & Veitch, V. (2024). *The Linear Representation Hypothesis and the Geometry of Large Language Models.* ICML.

REPRESENTATION ENGINEERING: DIFFERENCE OF MEANS

Finding the direction

WITH CREDIBILITY MARKERS

“Traditional vaccine development is a meticulous process, typically requiring 10–15 years…”

SAME CLAIM, NO GROUNDING

“It’s just common sense that a vaccine should take years to develop…”

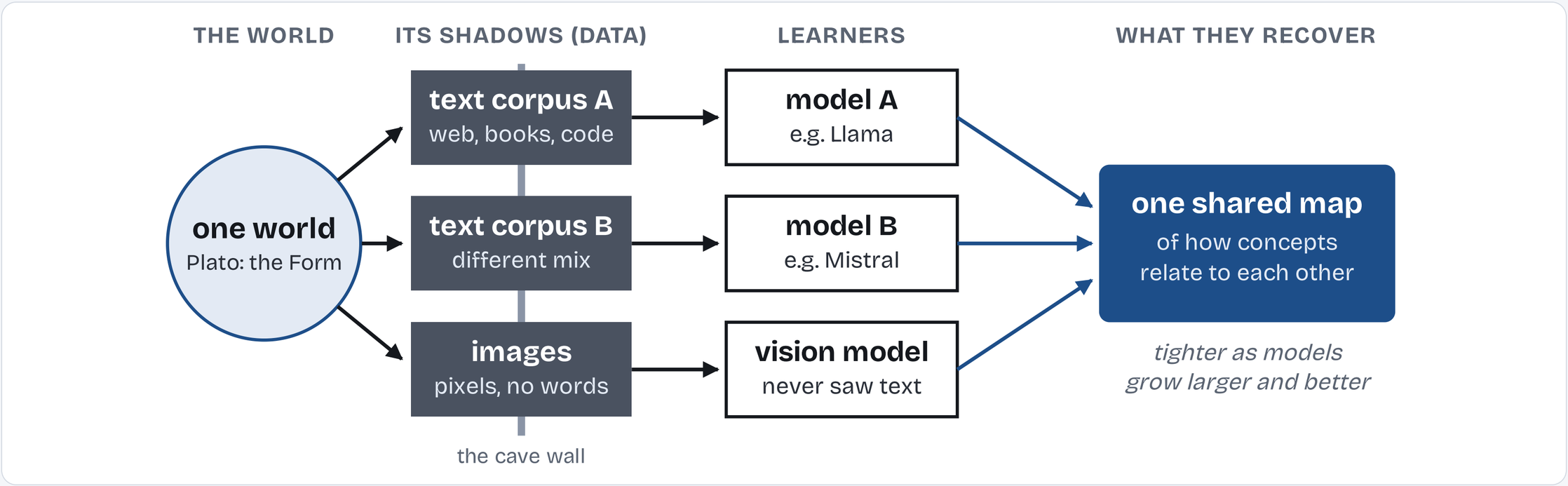
mean(with) − mean(without) = the concept vector, at every layer

17 concepts · about 250 pairs each · written by 14 different LLMs

Zou, A., et al. (2023). *Representation Engineering: A Top-Down Approach to AI Transparency.* arXiv:2310.01405. · Rimsky, N., et al. (2024). *Steering Llama 2 via Contrastive Activation Addition.* ACL.

THE PLATONIC REPRESENTATION HYPOTHESIS

Different shadows, one world?



What converges is relations, not coordinates.

Prediction: models that never shared a weight should agree on how concepts relate to each other.

Huh, M., Cheung, B., Wang, T., & Isola, P. (2024). *The Platonic Representation Hypothesis.* ICML.

THE QUESTION FOR THE POSTER

Do independently trained models agree?

If the hypothesis is right, models built by different teams on different text should agree on how their concepts relate to one another. I tested that across 35 open models, concept by concept and layer by layer.

If they do, is it the structure of language itself, or of the world it describes?

The results are on the poster. Come and talk it through. Code and reading list: github.com/jamesrahenry/ai-day-2026